

ALBERT HO

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EDUCATION

Bachelor of Science, Computer Science | University of California, San Diego
Awards: Summer URS Philip and Elizabeth Hiestand Scholarship

Expected: June 2026
GPA: 3.7

EXPERIENCES

Software Engineer Research Intern | UCSD Health (Hojun Li Lab)

October 2024 - Present

- Engineered a scalable Python-based bioinformatics pipeline for parsing BAM alignment files from Oxford Nanopore's Dorado platform, enabling efficient detection and quantification of A-to-I RNA editing events
- Developed a computationally optimized algorithm to calculate per-base modification frequencies, leveraging ONT's statistical models and vectorized Pandas/Numpy operations to minimize runtime and memory overhead
- Implemented automated genomic feature annotation using interval tree data structures, producing reproducible, research-ready datasets that support downstream analyses
- Engineered a reproducible, scalable workflow for more than 57,000 single-cell profiles using R, Seurat, and Python/Pandas, improving pipeline maintainability and computational efficiency

Data Science Intern | Lawrence Livermore National Laboratory | [Link](#)

July 2024 - August 2024

- Developed a production-ready machine learning pipeline for ventricular arrhythmia classification, benchmarking Logistic Regression, Random Forest, and deep learning models (CNNs and RNNs)
- Achieved 99% recall by designing and hyperparameter-tuning a sequence-to-sequence convolutional neural network, significantly improving diagnostic sensitivity
- Processed and labeled large datasets (over 16,000 samples) using Pandas dataframes for training and testing

PROJECTS

kumQAt (AI QA Automation Platform) | TypeScript, Next.js, FastAPI, Python, Gemini | [Link](#)

April 2026

- Led product direction and workflow design for an AI-powered QA platform converting natural language requirements into executable testing pipelines
- Architected a full-stack application with async orchestration enabling parallel test execution and real-time streaming via Server-Sent Events (SSE)
- Designed a multi-stage LLM pipeline (generation, execution, validation) producing structured results with triage-ready outputs
- Implemented autonomous browser-based testing with execution traces, screenshots, and persistent run history, supporting reruns and scheduled QA workflows

Hall Pass | TypeScript, React Native, Firebase, AWS S3, Expo | [Link](#)

January 2026 - Present

- Designed a full-stack social media platform for universities, managing the project lifecycle from UI design to deployment
- Implemented backend using Firebase for real-time messaging and AWS S3 for image storage, implementing private course-specific discussion spaces and verification
- Utilized AI-augmented development workflows to assist feature prototyping and optimize TypeScript codebases, while maintaining system logic and security protocols

Stock-em | Python, Flask, PyTorch, yfinance, Hugging Face, HTML, JS | [Link](#)

October 2025 - Present

- Developed a LSTM, RNN hybrid model to predict futures using historic stock prices and news sentiment scoring
- Incorporated a backtesting model to benchmark predicted values on existing data; displayed the model's buy, sell, and short calls, as well as outcomes such as profits, losses, SHARPE ratio, and return percentages
- Implemented a full stack web application capable of allowing users test the pipeline on different tickers and view returns

TECHNICAL SKILLS

Languages | Python, R, C/C++, Swift, TypeScript, HTML, CSS, JavaScript, Java

Technologies | Agile, React, Node.js, Tailwind CSS, Git, Github, VS Code, Firebase, AWS, PyCharm, XCode, scikit-learn, tensorflow, PyTorch, pandas, NumPy, Matplotlib, D3.js

Relevant Coursework | Data Structures and Algorithms, Operating Systems, AI: Probabilistic Models, Software Engineering, Engineering Probability and Statistics, Intro to Cryptography, Theory of Computation

INTERESTS

Machine Learning, Bioinformatics, Cybersecurity (Cryptography & Exploit Development), Playing Guitar/Piano, Global Travel